

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Notes and disclosures

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Notes on website version

This dissertation is available for viewing online at aastedet.github.io/dissertation/ (or by scanning the QR code in the lower right-hand corner of the title page). While the core contents are identical, the website may offer a more comfortable reading experience compared to the physical booklet for some readers. In particular, the website is recommended for readers with impaired vision and others who may be uncomfortable reading the small physical book.

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Preface

Motivation

During my work as a junior doctor in a general practice in Gellerup, a deprived suburb of Aarhus with a large migrant population, I was immediately struck by the high frequency of type 2 diabetes (T2D) and other chronic diseases among day-to-day consultations. I would soon learn that T2D held more surprises for me than just a high prevalence. After a few weeks, it seemed that much of what I had learned about the disease in medical school and studied in clinical guidelines did not apply in my new setting. Diabetes was not only more prevalent; the patients were also much younger, and their haemoglobin-A1c levels rarely came close to guideline targets despite our best efforts to intensify treatment. Sadly, these experiences would repeat themselves in my subsequent work with migrants as a general practitioner (GP) in health clinics of the Danish Red Cross at asylum residence centres and pre-removal detention centres.

As I reviewed the literature, I understood why medical school and guidelines had failed to prepare me for the challenges I faced as a clinician with T2D in migrants; the existing literature was inadequate. While increased prevalence of T2D appeared well-established, and migrants with T2D were also ascribed a higher mortality than their native counterparts, this provided little guidance for my day-to-day work. Contrarily, there was hardly any evidence on disparities in the time between diagnosis and death - the time when care from the GP is needed the most. So I was left to wonder what might cause this discordance between what I knew about T2D from books and guidelines, and what I encountered in migrants in the clinic.

*Are migrants with T2D more prone to under-treatment, and in which areas of care?
Are some migrant groups more prone to under-treatment than others?*

Knowledge on these clinical questions could enable GPs and healthcare planners to address disparities and improve care in migrants by prioritising and focussing care accordingly. While migrants are currently a younger demographic group than the rest of the population, it is likely that migrants will constitute an increasing proportion of the T2D population in the coming years as the group ages. Therefore, I was excited to explore these clinical questions as a researcher, hoping I could provide answers to

myself and fellow GPs that could lead to better care - and, ultimately, better health - in a vulnerable and challenging group of patients.

During my time as a PhD student, I quickly encountered the first of many challenges on the way to studying migrants with T2D in the Danish registers: there was no validated definition of T2D, nor a common consensus among researchers. This led to an expansion of the scope of the PhD project to develop and validate a tool to define T2D in the Danish registers, which allowed me to answer my research questions based on robust findings. This PhD project benefited greatly from open-source tools. In the spirit of open-source, the source code of the validated diabetes classifier was made available to other researchers in the *osdc* package for the *R* statistical programming language. As a final commitment to openness and accessibility, this dissertation was made available to a global public audience in website format.

Outline of the dissertation

Chapter 1 describes type 2 diabetes (T2D) and how migrants are at particular risk. It then outlines the context of migration and health care in Denmark and clinical guideline recommendations for T2D. Finally, it introduces the reader to identification of diabetes patients in Danish healthcare registers.

Chapter 2 states the overall aims of this PhD project and each individual study.

?@sec-methods describes the setting, data sources, methods and study designs used in the studies of this PhD project.

?@sec-results presents the main results of the studies.

?@sec-discussion-methods contains a discussion of the methods used and their potential impact on results.

?@sec-discussion-results discusses the main findings of each study and compares them to existing literature. Finally, the clinical implications of the results are discussed.

?@sec-conclusion presents the main conclusions.

?@sec-perspectives addresses the future perspectives raised by the dissertation and discusses the opportunities for research in the evolving research field.

The rest of the dissertation includes a list of **references**, English and Danish **summaries**, a **supplementary** validation analysis, and the three **papers** of the dissertation and their supplementary materials.

Papers in the dissertation

Study I

Isaksen AA, Sandbæk A, Bjerg L. Validation of Register-Based Diabetes Classifiers in Danish Data. *Clinical Epidemiology*. 2023;15:569-581. <https://doi.org/10.2147/CLEP.S407019>

Study II

Isaksen AA, Sandbæk A, Skriver MV, Andersen GS, Bjerg L (2023) Guideline-level monitoring, biomarker levels and pharmacological treatment in migrants and native Danes with type 2 diabetes: Population-wide analyses. *PLOS Global Public Health* 3(10): e0001277 <https://doi.org/10.1371/journal.pgph.0001277>

Study III

Isaksen, AA, Sandbæk, A, Skriver, MV, Bjerg, L. Glucose-lowering drug use in migrants and native Danes with type 2 diabetes: Disparities in combination therapy and drug types. *Diabetes Obesity and Metabolism*. 2023; 1-10. <https://doi.org/10.1111/dom.15230>

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Abbreviations

ACEI: Angiotensin-converting enzyme-inhibitors
APT: Antiplatelet therapy
ARB: Angiotensin receptor blockers
ATC: Anatomical Therapeutic Chemical (classification)
CI: Confidence interval
CVD: Cardiovascular disease
DKD: Diabetic kidney disease
DPP4i: Dipeptidyl peptidase-4 inhibitors
GP: General practitioner
GDM: Gestational diabetes mellitus
GLD: Glucose-lowering drugs
GLP1RA: Glucagon-like peptide-1 receptor agonists
HbA1c: Haemoglobin-A1c
LDL-C: Low-density lipoprotein cholesterol
LLD: Lipid-lowering drugs
NPV: Negative predictive value
OR: Odds ratio
OSDC: Open-Source Diabetes Classifier
PCOS: Polycystic ovary syndrome
PPV: Positive predictive value
RR: Relative risk
RSCD: Register of Selected Chronic Diseases
SGLT2i: Sodium glucose co-transporter type 2 inhibitors
T1D: Type 1 diabetes mellitus
T2D: Type 2 diabetes mellitus
UACR: Urine albumin-to-creatinine ratio

1. Introduction

1.1. Diabetes and cardiovascular disease

- Burden of diabetes
 - Type 1 diabetes
 - Type 2 diabetes
- Cardiovascular complications
- Heart failure in diabetes
- NT-proBNP as a biomarker

1.2. Glycemic markers and glucose monitoring

- HbA1c and its limitations
- Continuous glucose monitoring (CGM)
- Glycemic variability
 - Metrics of glycemic variability
 - Clinical challenges of glycemic variability (Think HbA1c and MBG and variability study)

1.3. Cardiovascular risk prediction in diabetes

- Traditional risk models
 - SCORE2-Diabetes risk score
- Limitations of conventional predictors

1.4. Machine learning

- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
 - Logistic regression
 - LSTM and CNNs
 - Foundation models
- Challenges and pitfalls of machine learning in healthcare
 - Overfitting and generalizability
 - Interpretability and explainability

1.5. Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Challenges in healthcare applications when using ML

2. Aims

2.1. Overall aims

The overall aim of this PhD dissertation is to provide robust knowledge on specific areas of T2D care where disparities between migrant groups and native Danes may be present.

2.2. Specific aims

- I. To develop a register-based classification of type 1 and type 2 diabetes that incorporates previous recommendations on data sources, and to make it available to other researchers.
- II. To validate register-based diabetes classifiers available to researchers in Denmark.
- III. To assess the prevalence of T2D in migrant minorities of Denmark, and to explore care disparities between migrant groups and native Danes in terms of monitoring, biomarker control and pharmacological treatment as recommended in the guidelines.
- IV. To evaluate disparities in the quality of glucose-lowering pharmacological treatment between migrants and native Danes in terms of combination therapy and drug types.

2.3. Outputs

Aim I was pursued in the creation of the *Open-Source Diabetes Classifier* (OSDC), made available in the open-source *R* package *osdc*.¹ Aims II, III, and IV were pursued in the respective papers I, II, and III of this dissertation.

3. Methods and Materials

In this chapter the three studies included in this PhD will be presented. The data will be described together with the chosen methods of the three studies and the different clinical variables, CGM-derived metrics and relevant cardiovascular outcomes will be explained.

3.1. Study Design Overview

- Ad table
 - Title
 - Design
 - Study population (Sample size, disease, cohort (CANCAN or T1DX)) ##
Study Populations Two datasets were used in this PhD (Study II and study III) which are presented below ### CANCAN The CANCAN data was collected in an observational case-control study from three hospital outpatient clinics in Denmark, Jutland. The aim of the CANCAN Study was to investigate if cardiac autonomic neuropathy (CAN) assessments, CGM and heart failure indicators could be used in risk stratification in people with T2D. Participants were included if they were adults (< 18 years old) and had T2D for at least a year. Exclusion criteria were atrial fibrillation or laser-treated eye disease within the past three months, were pregnant or lactating, had a life-threatening disease (expected survival < 1 year), or had cognitive disabilities rendering them unable to understand and give consent. Data were collected during a clinical examination, during which blood and urine samples were obtained, and a blinded CGM sensor was inserted. Blood samples to obtain NT-proBNP values were collected when the CGM sensors were returned, after the 2-week CGM data collection period had ended (Figure 3.2).

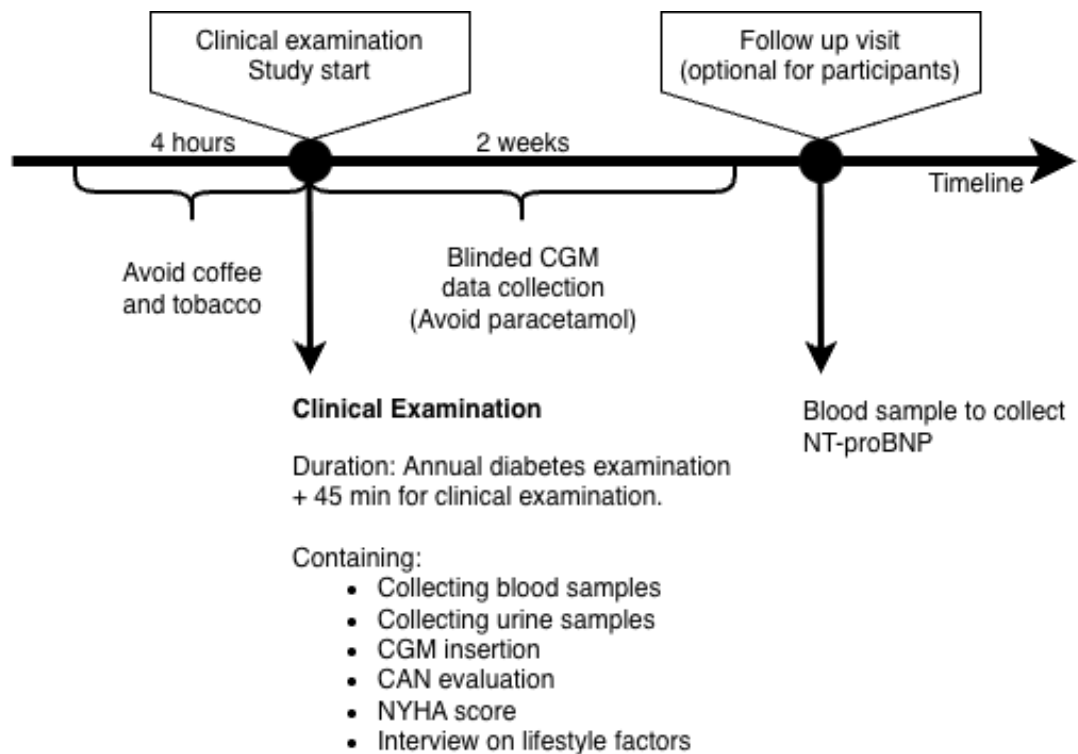


Figure 3.1.: Data collection timeline for the CANCAN Study

Figure 3.2.

3.1.1. T1DX

The T1DX racial differences dataset was collected by the T1D Exchange, a nonprofit organization, from diabetes centers in the United States (JAEB). The study focused on determining whether a racial difference exists in mean blood glucose and HbA1c levels between non-Hispanic Black and non-Hispanic White people with T1D. The study defined race and ethnicity through self-identified and self-reported ethnicity/race verification forms (Bergenstahl 2017). Further information on the definition of Hispanic, non-Hispanic, Black and White can be found in Bergenstahl et al.(Bergenstahl 2017). We, the authors, acknowledge that race and ethnicity are complex and not easily defined constructs, however we will use the word race moving forward since it aligns with the protocol and studies on the collection and analysis of the T1DX dataset (Kaneshiro2011, JAEB, Bergenstahl et al., 2017). Participants were grouped by age within each race. Between 8

and 17 years old and from 18 years and older. A blinded modified investigational version of Abbott Laboratories Diabetes Care's Flash Glucose Monitoring System (CGM) was used for 12 weeks. The study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the relevant institutional review boards. All participants provided written informed consent, with assent obtained where applicable (Bergenstahl 2017).

3.2. Definition of Predictors and Outcomes (Study I)

- Focus on the methods from the literature review (inclusion and exclusion criteria)
- Definition of cardiovascular risk (e.g., composite outcome, specific events)
- Definition of predictors <- What is CGM? MAGE from CGM or 7-point fingerpricks
- Present CGM-derived metrics
 - CGM preprocessing
- NT-proBNP as a biomarker of cardiovascular risk

3.3. Predictive Modeling and Machine Learning Framework

3.3.1. Learned representations of CGM data (Study II)

- CGM as a predictor of cardiovascular risk
- Foundation models
- Logistic regression
- Ridge regression

3.4. Bias in machine learning models (Study III)

- Fairness metrics
- Mitigation strategies -> Fine-tuning LSTM
- Evaluate bias when output is not binary (e.g., LMEM models)

4. Results

4.1. metrics used in the papers - Study 1 - Scoping review

- CGM metrics and CVD definitions
- Demographics
- Study designs / cohorts
- Diabetes types
- CGM metrics
- CVD risk factors
- Statistical methods used

4.2. Baseline characteristics table of CANCAN and T1DX cohorts (Study 2 and Study 3)

4.2.1. Representation learning of CGM data (Study 2)

4.2.2. Prediction of NT-proBNP (Study 2)

- Prediction NT-proBNP with different baseline models
- Predicting NT-proBNP using only age
- Predicting NT-proBNP using only CGM metrics
- Predicting NT-proBNP using SCORE2-Diabetes inputs and score together with other CGM data -> Supplementary material of paper 2

Copilot transition to next section: ### Bias in Machine Learning models (Study 3) *
Baseline characteristics of the T1DX cohort stratified by race status * Fairness metrics
of the LMEM model before and after fine-tuning

5. Discussion

6. Main findings

- Scoping review highlighted a chaotic landscape of wildly different study designs in the literature
- Mention other reviews, Yapanis + the other one that I forgot
- Representation learning of CGM data can capture complex patterns in glucose dynamics that are not captured by traditional CGM metrics
- Machine learning models can predict NT-proBNP levels using CGM data, however the performance of these models is modest and there is still room for improvement
- Bias is a concern in machine learning models, and fine-tuning on a subset of the data can help mitigate bias as measured by fairness metrics
- Grouped fairness is easier to evaluate and mitigate than individual fairness, but it is still an important step in ensuring that machine learning models are equitable and do not perpetuate existing biases in healthcare.

7. Machine learning vs traditional statistical methods

8. something about data and cross-sectional vs longitudinal data and how this affects the choice of methods

- Mention the collins protocol that creates a study focused on answer this question
- Mention the CANCAN dataset and the weird trend with people being better of the more comobiditie

9. Clinical implications

10. Strengths and limitations

11. Conclusions and future perspectives

12. Conclusion

- CGM data has the potential to provide valuable insights into cardiovascular risk in diabetes, but there are still many challenges to overcome in terms of study design, data quality, and machine learning methods.

13. Perspectives

References

1. Anders Aasted Isaksen. The Open-Source Diabetes Classifier [Internet]. {last accessed 7 Feb 2023}; Available from: <https://github.com/Aastedet/osdc>

English summary

Background: Type 2 diabetes (T2D) is an increasing burden on healthcare services due to changes in demography and lifestyle. Migrants are particularly vulnerable to T2D, having a higher prevalence of the disease compared to native populations in western Europe, with some evidence suggesting a higher risk of complications and mortality. Evidence on disparities in care that may contribute to this excess risk in migrants is sparse, but register-based studies are well-suited to explore this in a Danish context. However, these studies are vulnerable to biases due to the lack of a validated method to identify T2D cases in the general population.

Aims: This dissertation aimed to develop and validate a register-based classification of type 1 diabetes (T1D) and T2D, and utilise it to explore specific areas of monitoring, biomarker levels, and pharmacological treatment, where disparities in T2D care between migrant groups and native Danes may contribute to excess complication risk in migrants.

Methods: Three cross-sectional register-based studies were performed. Study I validated two register-based classifications of diabetes - the *Open-Source Diabetes Classifier* (OSDC), and the *Register of Selected Chronic Diseases* (RSCD) - in a population of survey respondents from Central Denmark Region (*Health In Central Denmark*), while the two subsequent studies examined T2D prevalence and care in nationwide populations, using native Danes as the reference group. Study II computed relative risk (RR) of prevalent T2D and non-fulfilment of eleven indicators corresponding to guideline-recommendations. Study III computed RR of using glucose-lowering drug (GLD) combination therapy, and RR of using each type of GLD.

Results: In study I, both register-based diabetes classifiers identified valid populations of T1D and T2D in a general population (positive predictive values from 87.5% to 94.4%, negative predictive values above 98.9%), although sensitivity in the OSDC was substantially higher compared to the RSCD in both T1D (77.3% vs. 70.0%) and in T2D (94.4% vs. 90.5%). Neither algorithm was able to accurately classify diabetes type in individuals with T1D onset after age 40, nor T2D onset before age 40. In study II, prevalence of T2D was higher in all migrants excluding the Europe-group (fully-adjusted RR from 0.96 [95% confidence interval: 0.94-0.99] (Europe group) to 4.04 [3.89-4.20] (Sri Lanka group)). Non-fulfilment of recommendations for care was common regardless of origin: in eight of the eleven indicators of care studied >25% of native Danes did not meet

guideline-recommendations. Apart from monitoring in the Sri Lanka group, migrants were at similar or higher risk of non-fulfilment than native Danes across all indicators of monitoring (crude RR from 0.64 [0.51-0.80] (hemoglobin-A1c (HbA1c) monitoring, Sri Lanka group) to 1.62 [1.36-1.95] (HbA1c monitoring, Somalia group)), HbA1c control (crude RR from 1.27 [1.24-1.30] (Middle East group) to 1.46 [1.41-1.52] (Pakistan group)), and low-density lipoprotein cholesterol (LDL-C) control (crude RR from 1.08 [1.03-1.14] (Former Yugoslavia group) to 1.78 [1.67-1.90] (Somalia group)), while no overall disparities were observed for the presence of pharmacological treatment (crude RR from 0.61 [0.46-0.80] (GLD, Sri Lanka group) to 1.67 [1.34-2.09] (GLD, Somalia group)). In study III, migrants were less likely to use GLD combination therapy than native Danes (fully adjusted RR from 0.77 [0.71-0.85] (Somalia group) to 1.00 [0.97-1.04] (Former Yugoslavia group)). Migrants were more likely to use oral GLD types (fully adjusted RR (use of any oral GLD) from 0.99 [0.97-1.01] (Europe group) to 1.09 [1.06-1.11] (Sri Lanka group)), but less likely to use injection-based GLD types such as insulins (fully adjusted RR from 0.66 [0.62-0.71] (Sri Lanka group) to 0.94 [0.89-0.99] (Europe group)) and, especially, glucagon-like peptide-1 receptor agonists (GLP1RA) (fully adjusted RR from 0.29 [0.22-0.39] (Somalia group) to 0.95 [0.89-1.01] (Europe group)).

Conclusion and perspectives This dissertation was able to accurately identify populations of T1D and T2D in register data from the general Danish population. Diabetes epidemiologists can rely on the OSDC and the RSCD to provide valid study populations for Danish register-based research, but researchers should be aware of limitations in cases with atypical age at onset and potential challenges due to evolving GLD indications in the future. This dissertation found room for improvement in T2D care in Denmark, and presented several disparities in T2D care between migrants and native Danes. These may contribute excess risk of complication and mortality in migrants, especially among migrants from Somalia, who received the poorest care of all groups overall. Overall, the large disparity in use of GLP1RA between migrants and native Danes may provide a target for future interventions to improve care and reduce complication risk among migrants with T2D.

Dansk resumé

Baggrund: Type 2 diabetes (T2D) er en stigende belastning på sundhedsvæsenet på grund af ændringer i demografi og livsstil. Indvandrere er særligt sårbare overfor T2D, idet de har en højere forekomst af sygdommen i forhold til indfødte befolkninger i Vesteuropa, ligesom nogle undersøgelser peger på en højere risiko for diabetiske komplikationer og død. Potentielle uligheder i behandlingen, der kan bidrage til den øgede risiko hos indvandrere med T2D, er dårligt belyst, men registerbaserede studier er velegnede til at undersøge dette i en dansk sammenhæng. Disse undersøgelser er imidlertid sårbare, idet der ikke foreligger nogen validerede metoder til at identificere T2D-tilfælde i den generelle danske befolkning.

Formål: Denne afhandling havde til formål at udvikle og validere en registerbaseret klassifikation af type 1 diabetes (T1D) og T2D, og anvende den til at undersøge specifikke områder i behandlingen af T2D - kontroller, biomarkør-niveauer og medicinsk behandling - hvor uligheder mellem indvandrergupper og indfødte danskere kan bidrage til øget risiko for komplikationer hos indvandrere.

Metoder: Som led i afhandlingen blev tre tværsnitstudier baseret på register-data gennemført. Studie I validerede to registerbaserede klassifikationer af diabetes - *Open-Source Diabetes Classifier* (OSDC) og *Register for Udvalgte Kroniske Sygdomme* (RUKS) - blandt spørgeskema-besvarelser fra personer i Region Midtjylland (*Helbred i Midt*), mens de to efterfølgende studier undersøgte forekomst og behandling af T2D i hele landet og sammenlignede indvandrergupper med indfødte danskere. Studie II beregnede relativ risiko (RR) for forekomst af T2D og for manglende opfyldelse af elleve indikatorer for behandlingskvalitet svarende til anbefalinger i gældende kliniske retningslinjer. Studie III beregnede RR for brug af kombinationsbehandling med blodsukker-nedsættende lægemidler (GLD) og RR for brug af hver type GLD.

Resultater: I studie I identificerede begge registerbaserede diabetesklassifikationer gyldige populationer af T1D og T2D i en bred befolkning (positive prædiktive værdier fra 87,5% til 94,4%, negative prædiktive værdier over 98,9%), dog med en væsentlig højere sensitivitet i OSDC sammenlignet med RUKS både i T1D (77,3% vs. 70,0%) og i T2D (94,4% vs. 90,5%). Ingen af algoritmerne var i stand til nøjagtigt at klassificere diabetestype hos personer med T1D debut efter 40-årsalderen eller T2D debut før 40-årsalderen. I studie II var forekomsten af T2D højere i alle indvandrergupper undtagen

Europa-gruppen (fuldt justeret RR fra 0,96 [95% konfidensinterval: 0,94-0,99] (Europa-gruppen) til 4,04 [3,89-4,20] (Sri Lanka-gruppen)). Manglende opfyldelse af de vejledende behandlingsanbefalinger var hyppig - uanset oprindelse: i otte ud af elleve indikatorer opfyldte >25% af indfødte danskere ikke anbefalingerne. Fraset rutinemæssig kontrolundersøgelser i Sri Lanka-gruppen, havde indvandrergupperne tilsvarende eller højere risiko end indfødte danskere for ikke at opfylde behandlingsanbefalinger på tværs af alle indikatorer for rutinemæssig kontrolundersøgelser (rå RR fra 0,64 [0,51-0,80] (hæmoglobin-A1c (HbA1c-undersøgelse, Sri Lanka-gruppen) til 1,62 [1,36-1,95] (HbA1c-undersøgelse, Somalia-gruppen)), blodsukterniveau (rå RR fra 1,27 [1,24-1,30] (Mellemøsten-gruppen) til 1,46 [1,41-1,52] (Pakistan-gruppen)) og kolesterolniveau (rå RR fra 1,08 [1,03-1,14] (Eksjugoslavien-gruppen) til 1,78 [1,67-1,90] (Somalia-gruppen)), mens der ikke sås nogen gennemgående forskelle for tilstedeværelse af medicinsk behandling (rå RR fra 0,61 [0,46-0,80] (GLD, Sri Lanka-gruppen) til 1,67 [1,34-2,09] (GLD, Somalia-gruppen)). I studie III var indvandrere mindre tilbøjelige til at få GLD kombinationsbehandling end indfødte danskere (fuldt justeret RR fra 0,77 [0,71-0,85] (Somalia-gruppen) til 1,00 [0,97-1,04] (Eksjugoslavien-gruppen)). Indvandrere var mere tilbøjelige til at benytte orale GLD typer (fuldt justeret RR (brug af ethvert oralt GLD) fra 0,99 [0,97-1,01] (Europa-gruppen) til 1,09 [1,06-1,11] (Sri Lanka-gruppen)), men mindre tilbøjelige til at benytte injektionsbaserede GLD typer så som insulin (fuldt justeret RR fra 0,66 [0,62-0,71] (Sri Lanka-gruppen) til 0,94 [0,89-0,99] (Europa-gruppen)) og især glucagon-like peptide-1 receptor agonister (GLP1RA) (fuldt justeret RR fra 0,29 [0,22-0,39] (Somalia-gruppen) til 0,95 [0,89-1,01] (Europa-gruppen)).

Konklusion og perspektiver: Denne ph.d.-afhandling kunne præcist identificere befolkninger med T1D og T2D i den brede danske befolkning ud fra registerdata. Diabetesforskere kan anvende OSDC og RUKS til at danne gyldige studiepopulationer ud fra de danske registre, men bør være opmærksomme på deres begrænsninger, herunder blandt patienter med atypisk alder for sygdomsdebut samt potentielle udfordringer som følge af udviklingen i brugen af GLD i fremtiden. Afhandlingen fandt plads til forbedringer i behandlingen af T2D i Danmark, og viste flere uligheder mellem indvandrere og indfødte danskere i behandlingen af T2D, som kan medføre øget risiko for komplikationer og død blandt indvandrere - særligt blandt indvandrere fra Somalia, som havde den laveste behandlingskvalitet af alle grupper. Den store forskel i brugen af GLP1RA mellem indvandrere og indfødte danskere kan muligvis være et mål for fremtidige interventioner til at øge behandlingskvaliteten og mindske komplikationsrisikoen blandt indvandrere med T2D.

A. Supplementary validation analyses

A.0.1. Validation of unspecified diabetes in the Health In Central Denmark survey and in the National Health Survey.

Validation analyses performed in the HICD study population from study I:

Validation analyses performed among respondents in the National Health Survey 2017 (individuals reporting diabetes as a current disease were considered diabetes cases).

Table A.1.: **Supplementary: Validation of unspecified diabetes in the HICD study population**

Table A.2.: The Open-Source Diabetes Classifier (OSDC)

OSDC	Survey: +DM	Survey: -DM	Total N
OSDC: +DM	2,512	221	2,733
OSDC: -DM	121	26,537	26,658
Total N	2,633	26,758	29,391
Sensitivity	0.954 (0.945, 0.962)		
Specificity	0.992 (0.991, 0.993)		
PPV	0.919 (0.908, 0.929)		
NPV	0.995 (0.995, 0.996)		

Table A.3.: The Register for Selected Chronic Diseases (RSCD)

RSCD	Survey: +DM	Survey: -DM	Total N
RSCD: +DM	2,422	122	2,544
RSCD: -DM	211	26,636	26,847
Total N	2,633	26,758	29,391
Sensitivity	0.920 (0.909, 0.930)		
Specificity	0.995 (0.995, 0.996)		
PPV	0.952 (0.943, 0.960)		
NPV	0.992 (0.991, 0.993)		

Table A.4.: **Supplementary: Validation of unspecified diabetes in the National Health Survey**

Table A.5.: The Open-Source Diabetes Classifier (OSDC)

OSDC	Survey: +DM	Survey: -DM	Total N
OSDC: +DM	1,630	258	1,888
OSDC: -DM	87	26,460	26,547
Total N	1,717	26,718	28,435
Sensitivity	0.95 (0.94, 0.96)		
Specificity	0.99 (0.99, 0.99)		
PPV	0.86 (0.85, 0.88)		
NPV	1.00 (1.00, 1.00)		

Table A.6.: The Register for Selected Chronic Diseases (RSCD)

RSCD	Survey: +DM	Survey: -DM	Total N
RSCD: +DM	1,546	184	1,730
RSCD: -DM	171	26,534	26,705
Total N	1,717	26,718	28,435
Sensitivity	0.90 (0.89, 0.91)		
Specificity	0.99 (0.99, 0.99)		
PPV	0.89 (0.88, 0.91)		
NPV	0.99 (0.99, 0.99)		

B. Full papers

B.1. Paper I

B.2. Paper II

B.3. Paper III

C. Addenda and errata

This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation (commit db88939).
- Errata: corrections to errors discovered since submission of the dissertation.

C.1. Addenda

- 5 May 2023:
 - Study I paper published. List of studies updated with citation and link to doi.
- 8 August 2023:
 - Study III paper published. List of studies updated with citation and link to doi.

C.2. Errata

- 2 May 2023:
 - On lines 5-10, page 81, section 5.1.5., the following excerpt on the interpretation of **?@fig-age-inclusion-by-year** states:
 - “Before 2016, when HbA1c data was not available from all regions of Denmark, a spike of inclusions is seen for women with T2D included at age 40 years. This spike is absent in the subsequent years, but the distribution is slightly skewed with a plateau in the age range 40-45 years. This is likely a residual effect of women in this age range having been prematurely included in the preceding years.”
 - * This statement misinterprets some elements of the figure, which are addressed below:

- * The density plots are smoothed, and spikes in single years are less visible compared to a histogram of observed counts (due to statistical disclosure requirements, this data could not be visualised using histograms). While the spike in included OSDC T2D-cases among women age 40 is indeed much lower in the years 2016-2018 than in the previous years, it is still present.
- * Most likely, the spike is caused by women with onset of T2D before the age of 40 years during a period of time without HbA1c data. These women may have initiated metformin treatment before HbA1c data became available, and if they achieved stable euglycaemia using only metformin, they would not be captured by the HbA1c inclusion criterion after the HbA1c data became available. Thus, their inclusion into the cohort would be delayed until the censoring of metformin purchases ceases at age 40 years, causing a spike at this age of inclusion. In the post-2016 data, women living in the Southern Denmark Region (which did not supply HbA1c data prior to 2015, compared to the other regions, which supplied data from 2011 onwards) are likely the major source of this spike.